

Molecular descriptor driven QSPR modeling of Papp, TEER and Efflux Ratio from Caco-2 cells using machine learning for various phytochemicals

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Abstract

BACKGROUND: The present study aimed to develop and validate quantitative structure–property relationship (QSPR) models for predicting permeability related bioavailability indicators including apparent permeability (Papp), trans-epithelial electrical resistance (TEER) and efflux ratio (ER) based on molecular descriptors ($n = 5003$) of 83 phytochemicals. A rigorous workflow integrating feature selection and 10 regression algorithms were employed to assess predictive performance.

RESULTS: Pearson correlation coefficient analysis revealed significant relationships among the three indicators. Among the models, a stacking ensemble, including CatBoost, LightGBM and Gradient Boosting as base learners and linear regression as the meta-model, achieved strong predictive accuracy for Papp and ER, at the same time as showing moderate but highly variable performance for TEER. SHapley Additive exPlanations (i.e. SHAP)-based feature importance analysis provided insight into key molecular descriptors associated with electronic, topological and branching.

CONCLUSION: These results demonstrate the utility of interpretability tools in constructing robust and explainable QSPR models for bioavailability prediction in various phytochemicals.

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Supporting information may be found in the online version of this article.

Keywords: phytochemicals; bioavailability; machine learning; QSPR; SHAP-based feature importance analysis

INTRODUCTION

Phytochemicals, comprising plant-derived secondary metabolites, are key ingredients in functional foods and pharmaceuticals as a result of their diverse biological effects, including anticancer and antimicrobial properties.^{1–6} The classifications of these phytochemicals include carotenoids, polyphenols (including phenolic acids and lignans), flavonoids, alkaloids and ginsenosides.^{6–8} Because of their therapeutic properties, dietary phytochemicals have been shown to reduce cancer risk and lower the incidence of chronic diseases, with epidemiological studies supporting their role in chemoprevention.^{5,9–12} For example, specific compounds such as *Moringa oleifera* thiocarbamate glycosides and ginseng extracts demonstrate significant clinical and preventive effects against tumors, hypertension and various human cancers.^{7,8,13–17}

Understanding phytochemical bioavailability is crucial for maximizing their therapeutic effects because molecular structure and food matrix interactions significantly influence absorption.¹⁸ Bioavailability is commonly assessed using Lipinski's Rule of Five,

which predicts that specific structural constraints (e.g. hydrogen bond donors/acceptors, molecular weight and CLogP) lower absorption or permeability.^{19,20} Additionally, structural differences and cell membrane interactions govern cellular uptake, which acts as a key mechanism for biological activity.^{21–23} Furthermore, considering the comprehensive processes of absorption

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(A), distribution (D), metabolism (M) and excretion (E) is essential for pharmaceutical advancements.²⁴ Ultimately, because bioavailability determines the extent and rate at which bioactive compounds reach systemic circulation, it plays a critical role in determining their overall therapeutic efficacy.^{20,25}

Previous studies have utilized *in vitro* models, such as the Caco-2 cell system, to characterize the intestinal metabolism of phytochemicals, particularly their intestinal permeability, transport and membrane absorption, and these models have been used to predict the bioavailability of bioactive compounds.^{26–28} Building on this, the Caco-2 cell model has been widely recognized for its ability to provide critical insights into not only the intestinal absorption of bioactive compounds, but also the permeability and bioavailability of phytochemicals, serving as a fundamental step in the development of novel drugs and functional health ingredients.^{26,29} Despite its valuable contributions to drug discovery and functional ingredient research, the Caco-2 cell model has several limitations.^{30–33} One major drawback is the high cost and resource burden associated with its maintenance, particularly the 21-day differentiation period required for full cellular maturation.^{30–32} Permeability studies using this model require advanced analytical techniques, such as liquid chromatography-mass spectrometry or HPLC, further increasing financial and technical demands.³³

Recent studies have applied the quantitative structure–property relationship (QSPR) model to enhance the efficiency of analyzing biochemical properties, such as bioavailability, using Caco-2 cells.^{34,35} The QSPR model is a machine learning-based approach that correlates experimentally derived bioavailability indicators apparent permeability (Papp), transepithelial electrical resistance (TEER) and efflux ratio (ER) with the structural, physical, and chemical properties of phytochemicals to predict their bioavailability.^{25,34,36,37} Although most previous studies have focused solely on developing predictive models for Papp, they often overlooked other key parameters for bioavailability, such as TEER, which reflects cellular integrity, and ER, which determines whether a compound is predominantly absorbed or expelled.^{35,37}

To address this gap, the present study aimed to develop explainable machine learning models capable of accurately predicting key permeability related indicators Papp, TEER and ER based on molecular descriptors of 83 phytochemicals. We also aimed to establish a robust QSPR framework that enables the early identification of phytochemicals with favorable intestinal absorption characteristics by integrating feature selection, regression algorithms and ensemble modeling.

MATERIALS AND METHODS

Chemicals and reagents

The analytical standards of (–)-cytisine, (–)-dihydroquinine, (+)-nootkatone, (+)-taxifolin, (E)-3-(4-chlorophenyl)acrylic acid, (E)-cinnamyl alcohol, 20(R)-ginsenoside Rg2, 20(R)-ginsenoside Rh1, 20(S)-ginsenoside Rg3, 20-O-glucoginsenoside Rf, 3-(3,4-dihydroxyphenyl)-L-alanine, 3,5-dimethoxy-4-hydroxycinnamic acid, aegeline, bacitracin, boldine, carminic acid (natural dye), cepharanthine, cholecalciferol, colchicine, compound K, cycloheximide, ergosterol, galantamine hydrobromide, ginsenoside F1, ginsenoside F2, ginsenoside F3, ginsenoside F5, ginsenoside Rb1, ginsenoside Rb2, ginsenoside Rb3, ginsenoside Rc, ginsenoside Re, ginsenoside Rf, ginsenoside Rg1, ginsenoside Rg2, ginsenoside Rg4, ginsenoside Rg5, ginsenoside Rh1, ginsenoside Rk3, gypenoside L, gypenoside XLIX, harmol, indican, jatrorrhizine chloride, lanatoside C, LDN-22684, L-tetrahydropalmatine,

lycorine, mangiferin, notoginsenoside Ft1, notoginsenoside R1, notoginsenoside R2, oleuropein, oridonine, paeoniflorin, phenazine methosulfate, piperine, piperlongumine, protopanaxatriol, reserpine, rhaponticin, rosmarinic acid, rutaecarpine, santonin, scopolamine hydrobromide trihydrate, sinigrin, swertiamarin, tabersonine, tanshinone IIA, tetrahydropalmatine, tetrandrine, trans-cinnamic acid, trans-*m*-coumaric acid, trans-*p*-coumaric acid, triamcinolone, trigonelline hydrochloride, trioxsalen, vina-ginsenoside R4, vincamine, wogonin, wogonoside, xanthotoxin, and yohimbine hydrochloride were obtained from TCI Chemical (Tokyo, Tapan).

Bioavailability indicators through permeability experiments

Permeability data for 83 polyphenols were obtained from *in vitro* experiments using differentiated Caco-2 monolayers previously conducted by our group. Permeability experiments were initiated 21 days post-seeding to confirm the full differentiation of the Caco-2 monolayers, which were selected based on an initial trans-epithelial electrical resistance (TEER) value exceeding $300 \Omega \text{ cm}^{-2}$ to ensure the integrity of the epithelial cell barrier. TEER measurements were conducted using the Millicell ERS-2 system (Millipore, Bedford, MA, USA). To assess bi-directional permeability, a diluted standard solution of polyphenols ($100 \mu\text{g mL}^{-1}$) was added to either the apical (AP, 500 μL) or basal (BL, 1500 μL) compartments. After incubation for 2 h, 500 μL of the transport medium was collected from both sides, stored at 4 °C, and subsequently analyzed by HPLC. The 2-h incubation time was selected based on previously established *in vitro* digestion models.^{38–40} TEER values were recorded at 15, 45, 90 and 120 min, and the data were expressed as a percentage of the baseline (0 min) using:

$$\text{TEER (\% of 0 min)} = \frac{\text{TEER value at each time point}}{\text{TEER value at 0 min}} \times 100 \quad (1)$$

The apparent permeability coefficients (Papp) and efflux ratio (ER) were calculated based on:

$$\text{Papp} = \frac{dQ}{dt} \times \frac{1}{C_0 \times A} \quad (2)$$

where (dQ/dt) represents the rate of polyphenol transport to the receiver side, expressed as the amount of substance transported per unit time after 120 min of incubation, C_0 is the initial concentration of polyphenols in the donor compartment and A is the membrane surface area (1.12 cm^2).

$$\text{Efflux ratio} = \frac{\text{Papp (BL} \rightarrow \text{AP)}}{\text{Papp (AP} \rightarrow \text{BL)}} \quad (3)$$

An ER greater than 2 suggests that the compounds may undergo active efflux. All experiments were performed in triplicate, resulting in a dataset of 249 samples, which was used to develop and validate predictive models for bioavailability indicators.

Descriptors selection, data preprocessing and QSPR model development

The overall step-by-step workflow for descriptor selection, model evaluation and feature refinement is illustrated in Fig. 1. First, to predict bioavailability indicators (Papp, TEER and ER) for 83 phytochemicals, molecular descriptors were generated from isomeric SMILES (i.e. simplified molecular input line entry system) retrieved

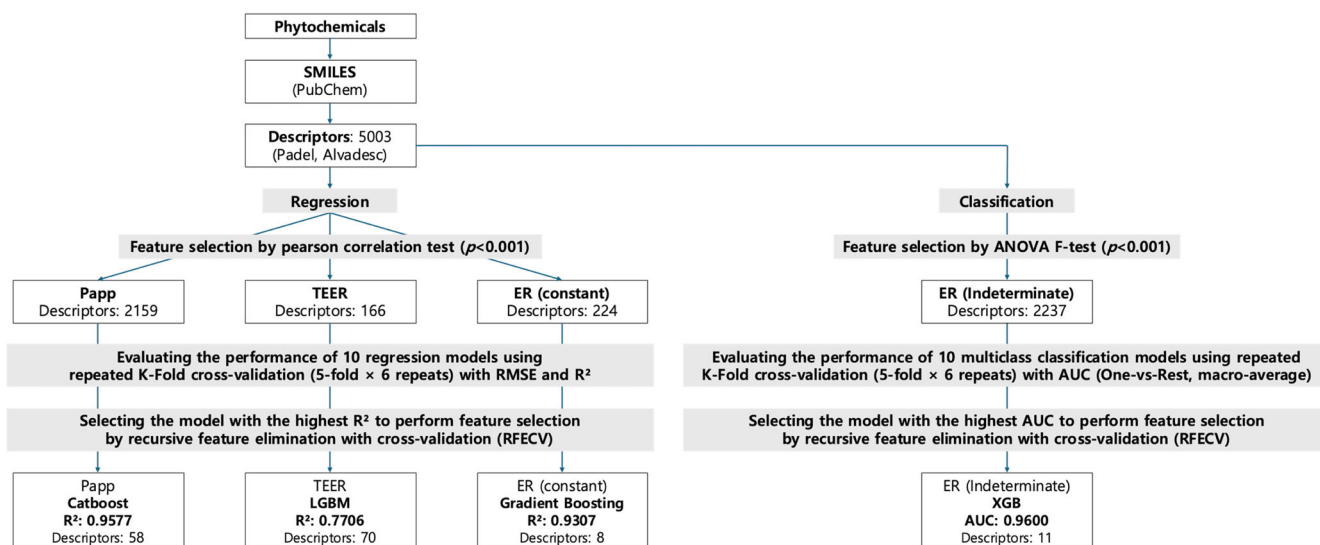


Figure 1. Feature selection and model evaluation workflow for regression and classification.

from PubChem. Descriptor calculation was performed using PaDEL-Descriptor⁴¹ and alvaDesc,⁴² yielding a comprehensive set of structural and physicochemical variables. Second, to reduce dimensionality and retain meaningful predictors, descriptors with statistically significant correlations to each indicator ($P < 0.001$) were selected based on Pearson correlation analysis. StandardScaler from the sklearn library⁴³ (Python, version 3.11.5; <https://www.python.org>) was applied to normalize descriptor values, with the scaler fit only to the training data to prevent data leakage. Third, 10 machine learning models, including CatBoost⁴⁴, random forest, XGBoost (XGB)⁴⁵, LightGBM (LGBM)⁴⁶, gradient boosting, linear regression, Bayesian ridge, bagging and support vector machines (SVM) with linear and RBF kernels, were evaluated. Model training and validation were conducted using repeated K -fold cross-validation ($n_splits = 5$, $n_repeats = 6$) to ensure robustness. The model achieving the highest R^2 on the test set was selected for further optimization. Finally, recursive feature elimination with cross-validation (RFECV) was applied to identify the most relevant descriptors. Final model performance was assessed on both training and test sets using normalized root mean squared error (NRMSE) and R^2 . Metrics reported include NRMSE and R^2 for both training ($NRMSE_{Train}$, R^2_{Train}) and test sets ($NRMSE_{Test}$, R^2_{Test}), providing a comprehensive evaluation of model accuracy and generalizability.

Construction of a multi-output stacking ensemble model for bioavailability (BA) indicators

Based on the results of the Pearson correlation coefficient (PCC) analysis, which revealed significant correlations among Papp, TEER and ER, a multi-output stacking ensemble model was constructed to leverage these interdependencies for improved prediction performance. Three distinct feature sets were initially selected for each target variable using RFECV. These were subsequently merged to create a unified input space (union feature set), which served as the shared input for three independent base regression models: CatBoost, LGBM and gradient boosting. Each base model was trained to simultaneously predict Papp, TEER and ER. The predictions from the base models (3 outputs $\times$ 3 models) were concatenated to form a stacked feature matrix of size $N \times 9$. The stacked predictions were then passed to a meta-

learner, implemented as a multi-output linear regression model (multi-output regressor with linear regression). This meta-model combined the base models' outputs to produce the final predictions for each of the three bioavailability indicators. Model training and validation were carried out using 5-fold cross-validation with six repetitions (repeated K -fold, $n_splits = 5$, $n_repeats = 6$, $random_state = 42$). Performance was evaluated based on the NRMSE and R^2 for each target variable.

Statistical analysis

All data are expressed as the mean $\pm$ SD in triplicate. Statistically significant differences among the samples were measured by applying one-way analysis of variance using Prism, version 6.01 (GraphPad Software Inc., San Diego, CA, USA). $P < 0.05$ was considered statistically significant and a 95% confidence level was determined for the post-hoc Tukey's test. The normality of the data was assessed using the Shapiro–Wilk test ($P > 0.05$). Subsequently, the PCC was employed to quantify linear relationships between variables. Statistical analyses were conducted using R, version 4.3.2 (R Foundation, Vienna, Austria).

RESULTS

Data processing and feature selection

Permeability of 83 phytochemicals experimentally assessed using Caco-2 cell assays yielded 83 data points in triplicate for each parameter such as Papp, TEER and ER (see Supporting information, Table S1). The isomeric SMILES of 83 phytochemicals, which encode stereochemistry, chemical structure and physicochemical properties, were utilized to generate 5659 molecular descriptors, comprising 1444 from PaDEL-Descriptor and 4215 from alvaDesc (data not shown). Duplicate descriptors present in both programs were removed to eliminate redundancy, resulting in a final set of 5003 unique molecular descriptors (Fig. 1). A systematic feature selection and model evaluation pipeline was subsequently applied. Pearson correlation analysis ($P < 0.001$) was used to filter relevant descriptors, yielding 2159 descriptors for Papp, 166 descriptors for TEER and 224 descriptors for ER. The selected descriptors were subsequently evaluated the performance of 10 regression models: CatBoost, random forest, XGB, LGBM,

bagging, gradient boosting, linear regression, SVM (linear), SVM (RBF) and Bayesian ridge (Table 1). Repeated K -fold cross-validation (5 folds $\times$ 6 repeats) was used to compute the NRMSE and coefficient of determination (R^2). Among the 10 models evaluated, performance varied significantly across the three permeability indicators, with tree-based ensemble algorithms generally outperforming linear and distance-based models (Table 1). For Papp, CatBoost ($R^2_{\text{Test}} = 0.9367$) and gradient boosting ($R^2_{\text{Test}} = 0.9237$) demonstrated superior generalization capability with minimal prediction error, suggesting robust modeling of non-linear molecular relationships. LGBM also exhibited favorable and stable performance. By contrast, models such as XGB, SVM and linear regression exhibited lower prediction performance, reflecting their limitations in capturing complex structure-permeability relationships. In the case of TEER, LGBM yielded the best predictive performance ($R^2_{\text{Test}} = 0.6797$). However, a noticeable gap between training and test R^2 values ($\Delta R^2 \approx 0.1739$) suggested moderate overfitting, implying a higher difficulty in generalizing this specific biological endpoint. For ER, gradient boosting achieved the highest test performance ($R^2_{\text{Test}} = 0.9306$), closely followed by CatBoost and XGB. The consistency between training and test metrics for these ensemble models indicated robust learning with minimal overfitting, whereas linear regression demonstrated a limited ability to capture the complex efflux dynamics. At the same time, RFECV was applied to refine the feature set, resulting in identifying 58 key descriptors for Papp, 70 for TEER and 8 for ER, showing the highest R^2 of 0.9577, 0.7706 and 0.9307 with CatBoost, LGBM and gradient boosting, respectively (Fig. 1). This approach enhanced the predictive performance of permeability modeling by improving feature selection efficiency and contributing to more reliable bioavailability assessments.

However, during ER data preprocessing, some values were either undefined or extreme, making them unsuitable for conventional regression modeling. As a result of the mathematical definition of the efflux ratio as the quotient of Papp (BL $\rightarrow$ AP) divided by Papp (AP $\rightarrow$ BL) [Eqn (3)], certain data points resulted in undefined or extreme values. Specifically, null efflux ratios occurred in cases where both Papp (AP $\rightarrow$ BL) and Papp (BL $\rightarrow$ AP) were zero, resulting in a mathematically undefined 0/0 condition. These cases were likely a result of compounds with extremely low permeability or poor solubility, resulting in negligible transport in both directions. On the other hand, infinite efflux ratios were observed when the denominator (Papp (AP $\rightarrow$ BL)) was zero, whereas the numerator (Papp (BL $\rightarrow$ AP)) remained non-zero, suggesting strong unidirectional efflux or selective transport through the membrane. Because such null and infinite values cannot be appropriately handled by standard regression models, only constant ER values (finite values) were used for regression-based prediction. To account for the excluded indeterminate ER cases, a separate classification strategy was implemented (Fig. 1). From the original 5003 molecular descriptors, an ANOVA F -test ($P < 0.001$) was applied to select 1815 features significantly associated with the categorical characteristics of efflux behavior. The ER values were categorized into three distinct classes, constant, infinite and null, representing finite, unidirectional, and mathematically undefined efflux, respectively. The classification performance of 10 machine learning models (CatBoost, random forest, XGB, LGBM, naive Bayes, bagging, gradient boosting, linear regression, SVM (linear) and SVM (RBF)) was evaluated using repeated K -fold cross-validation (5 folds $\times$ 6 repeats), with macro-averaged area under the curve (AUC) (one-vs-rest) as the evaluation metric. Among these, CatBoost demonstrated the

highest classification performance (macro-AUC = 0.8759) and was selected for further refinement using RFECV, which reduced the feature set to 22 informative descriptors. To further examine the class-wise discrimination, Receiver operating characteristic (ROC) curves were generated for each ER class using a one-vs-rest approach (Fig. 2). The Null class, corresponding to compounds with undefined ER (0/0), yielded the highest AUC of 0.9495, followed by the constant class (AUC = 0.9458). The infinite class, indicative of strong unidirectional efflux, showed a comparatively lower AUC of 0.7324, reflecting greater heterogeneity in its molecular features. These results supported the notion that, despite the absence of precise numerical ER values, efflux-related behaviors could still be predicted based on structural information through categorical classification. The high AUCs for null and constant classes suggested consistent molecular patterns associated with negligible or balanced transport, whereas the lower performance in the Infinite class implied possible involvement of diverse transporter-specific interactions. This classification pipeline enabled the inclusion of structurally atypical compounds that were not suitable for regression-based analysis. Together with the regression model built on finite ER values, this dual modeling strategy enhanced both the robustness and applicability of the overall efflux prediction framework.

Integrated multi-output stacking model for predicting bioavailability indicators

The interrelationship among Papp, TEER and ER was assessed using PCC to evaluate their potential interdependence (Fig. 3). It was found that the correlations were generally weak with having statistically significant ($P < 0.05$). In detail, Papp exhibited a weak negative correlation with ER ($r = -0.17$, $P < 0.05$) and a positive correlation with TEER ($r = 0.27$, $P < 0.001$), and TEER was negatively correlated with ER ($r = -0.25$, $P < 0.01$). Despite modest coefficients, these findings imply that the three permeability indicators are not entirely independent and may influence each other through subtle biological interactions. These relationships were visualized with scatter plots incorporating linear regression lines and 95% confidence ellipses, highlighting the distribution patterns across all 83 phytochemical samples.

To enhance the predictive accuracy of permeability-related indices, we implemented a stacking ensemble model integrating three tree-based regressors: CatBoost, LGBM and gradient boosting (Fig. 4). Each base model was trained using a common union feature set to simultaneously predict the three target outputs: Papp, TEER and ER. The individual predictions from each model (3 models $\times$ 3 outputs = 9 values per instance) were assembled into a stacked output matrix of shape $N \times 9$, which served as the input for the meta learner. A multi-output linear regression model was employed as the meta model to integrate base learners' predictions and produce the final ensemble output for Papp, TEER and ER. Model performance was evaluated using 5-fold cross-validation repeated six times (repeated K -fold, $n = 5$, repeats = 6) and both training and test set results were averaged over 30 iterations. The integrated stacking ensemble model, which combined CatBoost, LGBM and gradient boosting as base learners with linear regression as a meta-learner, demonstrated varying predictive capabilities across the three permeability indicators (Table 2). The model achieved the highest accuracy for Papp ($R^2_{\text{Train}} = 0.9862 \pm 0.0030$, $R^2_{\text{Test}} = 0.9550 \pm 0.0399$), with minimal discrepancy between training and testing performance, indicating excellent generalization without evidence of overfitting. Similarly, for ER, the ensemble model yielded robust predictive

Table 1. Evaluation of the performance of 10 regression models: performance metrics derived from cross-validation on the training and testing sets, as well as hyperparameters.

Model	Papp				TEER			
	NRMSE _{Train}	R ² _{Train}	NRMSE _{Test}	R ² _{Test}	NRMSE _{Train}	R ² _{Train}	NRMSE _{Test}	R ² _{Test}
CatBoost	0.0166 ± 0.0022	0.9870 ± 0.0027	0.0330 ± 0.0214	0.9367 ± 0.0696	0.0442 ± 0.0059	0.8700 ± 0.0320	0.0651 ± 0.0204	0.6370 ± 0.3257
Random forest	0.0222 ± 0.0020	0.9768 ± 0.0042	0.0440 ± 0.0203	0.8884 ± 0.0989	0.0447 ± 0.0054	0.8673 ± 0.0299	0.0654 ± 0.0170	0.6539 ± 0.2444
XGBoost	0.1463 ± 0.0085	−0.0000 ± 0.0000	0.1435 ± 0.0319	−0.0299 ± 0.0736	0.0426 ± 0.0061	0.8789 ± 0.0311	0.0660 ± 0.0215	0.6248 ± 0.3447
LGBM	0.0244 ± 0.0023	0.9716 ± 0.0069	0.0375 ± 0.0220	0.9225 ± 0.0669	0.0472 ± 0.0040	0.8536 ± 0.0270	0.0657 ± 0.0149	0.6797 ± 0.1422
Bayesian ridge	0.0156 ± 0.0021	0.9886 ± 0.0023	0.0390 ± 0.0329	0.8907 ± 0.1884	0.0628 ± 0.0068	0.7384 ± 0.0585	0.0835 ± 0.0154	0.4336 ± 0.3429
Bagging	0.0221 ± 0.0020	0.9770 ± 0.0042	0.0439 ± 0.0203	0.8889 ± 0.1007	0.0447 ± 0.0054	0.8675 ± 0.0300	0.0655 ± 0.0171	0.6517 ± 0.2490
Gradient boosting	0.0155 ± 0.0021	0.9886 ± 0.0023	0.0328 ± 0.0202	0.9237 ± 0.1354	0.0432 ± 0.0061	0.8758 ± 0.0315	0.0654 ± 0.0211	0.6285 ± 0.339
Linear regression	0.0155 ± 0.0022	0.9887 ± 0.0025	3.0269 × 10 ⁹	−2.092 × 10 ²¹	0.0431 ± 0.0063	0.8763 ± 0.0325	2.6057 × 10 ¹⁰	−1.9292 × 10 ²³
SVM (linear)	0.3829 ± 0.0389	−5.8764 ± 1.0350	0.3844 ± 0.0359	−8.5074 ± 7.5912	0.0799 ± 0.0062	0.5836 ± 0.0489	0.0905 ± 0.0195	0.4076 ± 0.1591
SVM (RBF)	0.3829 ± 0.0389	−5.8764 ± 1.0350	0.3844 ± 0.0359	−8.5074 ± 7.5912	0.1216 ± 0.0070	0.0399 ± 0.0110	0.1193 ± 0.0291	0.0215 ± 0.0371

Model	ER				Hyper parameters			
	NRMSE _{Train}	R ² _{Train}	NRMSE _{Test}	R ² _{Test}	iterations	n_estimators	random_state	random_seed
CatBoost	0.0347 ± 0.0017	0.9808 ± 0.0022	0.0629 ± 0.0137	0.9273 ± 0.0401	100	100	42	42
Random forest	0.0430 ± 0.0028	0.9703 ± 0.0055	0.0786 ± 0.0181	0.8928 ± 0.0416	100	100	42	42
XGBoost	0.0334 ± 0.0017	0.9822 ± 0.0019	0.0624 ± 0.0141	0.9268 ± 0.0444	100	100	42	42
LGBM	0.0394 ± 0.0028	0.9750 ± 0.0049	0.0741 ± 0.0252	0.8976 ± 0.0817	100	100	42	42
Bayesian ridge	0.0346 ± 0.0019	0.9809 ± 0.0022	0.0928 ± 0.0537	0.8025 ± 0.2681	100	100	42	42
Bagging	0.0432 ± 0.0029	0.9700 ± 0.0057	0.0787 ± 0.0188	0.8921 ± 0.0435	100	100	42	42
Gradient boosting	0.0335 ± 0.0017	0.9820 ± 0.0019	0.0613 ± 0.0129	0.9306 ± 0.0379	100	100	42	42
Linear regression	0.0356 ± 0.0061	0.9790 ± 0.0101	1.7542 × 10 ¹⁰	−2.0699 × 10 ²²	100	100	42	42
SVM (linear)	0.0429 ± 0.0016	0.9707 ± 0.0026	0.0918 ± 0.0448	0.8198 ± 0.2099	100	100	42	42
SVM (RBF)	0.0877 ± 0.0052	0.8775 ± 0.0126	0.1143 ± 0.0459	0.7682 ± 0.1501	100	100	42	42

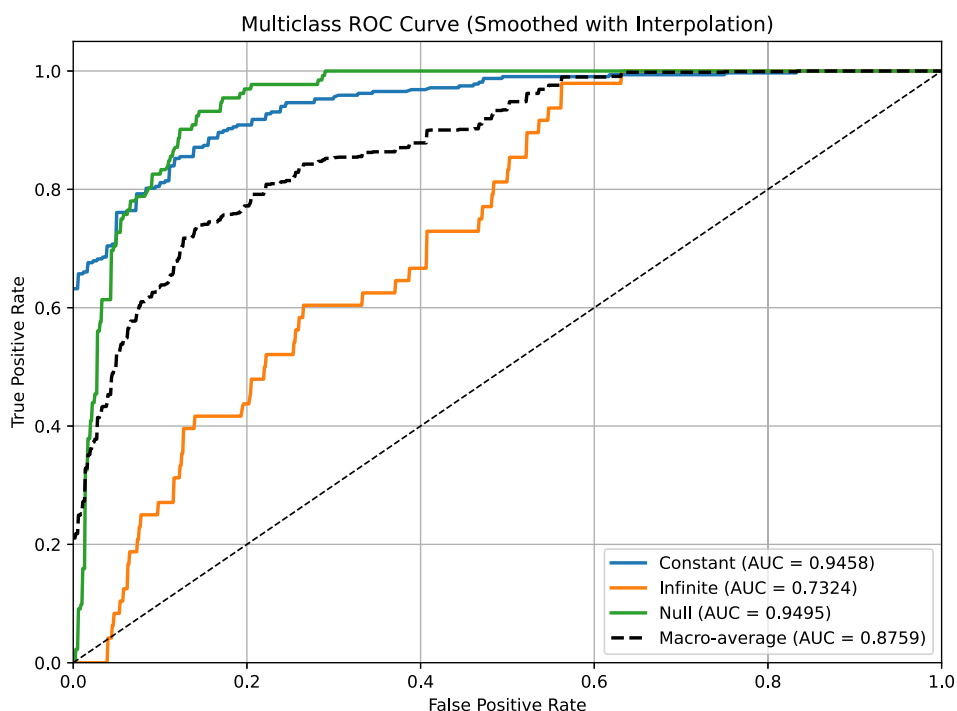


Figure 2. Multiclass ROC curves generated using a one-vs-rest approach for the classification of efflux ratio types: null, infinite and constant.

performance ($R^2_{\text{Train}} = 0.9821 \pm 0.0021$, $R^2_{\text{Test}} = 0.9289 \pm 0.0602$), effectively capturing efflux dynamics across structurally diverse phytochemicals. The close alignment of R^2 values suggested strong generalization capacity. By contrast, the model's performance for TEER was moderate ($R^2_{\text{Train}} = 0.8764 \pm 0.0365$, $R^2_{\text{Test}} = 0.5435 \pm 0.4556$). The notable decline in R^2 and the exceptionally large standard deviation observed for TEER predictions suggest possible overfitting or substantial biological variability. These outcomes likely reflect the inherently complex and dynamic nature of tight junction integrity and trans-epithelial ionic gradients, which may not be adequately represented by molecular descriptors alone. Moreover, the integration of heterogeneous, task-specific feature sets in the stacking ensemble may have introduced additional noise, further contributing to the instability of TEER predictions. As a result, the ensemble model failed to provide consistently reliable outputs for TEER, highlighting limitations in its applicability to biologically intricate endpoints. Notably, although the final feature sets for Papp, TEER and ER were non-overlapping, the stacking model was trained using a unified matrix composed of the union of all selected descriptors. For Papp and ER, this integration preserved relevant predictive information and did not degrade model performance, as evidenced by their consistently high R^2 values in both training and test sets. By contrast, TEER prediction remained highly variable and poorly generalizable, likely as a result of its greater biological sensitivity to factors not captured by structural descriptors. Given this instability, further refinement of the ensemble model for TEER was not pursued. These findings underscore the robustness of the stacking ensemble approach for permeability related indicators such as Papp and ER, as well as its limitations in modeling endpoints governed by complex biological processes, such as TEER.

Meaning of features

Figure 5 exhibits SHAP (i.e. SHapley Additive exPlanations)⁴⁶ based feature importance profiles for the prediction of three permeability indicators with selected molecular descriptors such as

58, 70 and 8 for Papp, TEER and ER, respectively. These descriptors collectively captured molecular electronic distribution (minaasC and VE3sign_D), topological and branching patterns (MDEC-33 and maxsssCH) and specific physicochemical properties such as conformational flexibility and functional group presence (nRCO and RotBtFrac). For Papp prediction, descriptor such as minaasC (minimum atom-type E-state for carbon atoms) emerged as contributed the most influential factor based on SHAP analysis (Fig. 5(A)), suggesting that the presence of carbon atoms with low electronic energy critically modulated the ability of molecules to traverse cellular membranes. This finding emphasized that atomic-level electronic environments, rather than molecular size alone, played a pivotal role in passive permeability. VE3sign_D, a descriptor related to the logarithmic coefficient sum of the last eigenvector from the topological distance matrix, ranked second in importance. This indicated that global topological organization and electronic delocalization within a molecule substantially impacted its membrane transport capability. Eta_D_beta, associated with the eta electronic feature balance, ranked third, highlighting that a fine balance of electronic features strongly contributed to permeability modulation. These top-ranked descriptors suggest that electronic distribution and topological structure were key determinants of membrane permeability in the phytochemicals. In addition, other features (see Supporting information, Table S2) that have an impact, including SpMin4_Bhm, representing the smallest absolute eigenvalue of the Burden matrix weighted by relative mass, and qpmax, indicating maximum positive charge, were also identified as important descriptors. And spectral moments such as SM04_EA(ed), SM05_EA(ed), SM08_EA and SM13_EA further reinforced the importance of local atomic environments and bonding patterns. Molecular walk counts (MWC07, MWC10) underscored the role of internal connectivity in influencing permeability pathways. Additionally, descriptors related to polarity and mass distribution, such as GATS5v (Geary autocorrelation weighted by van der

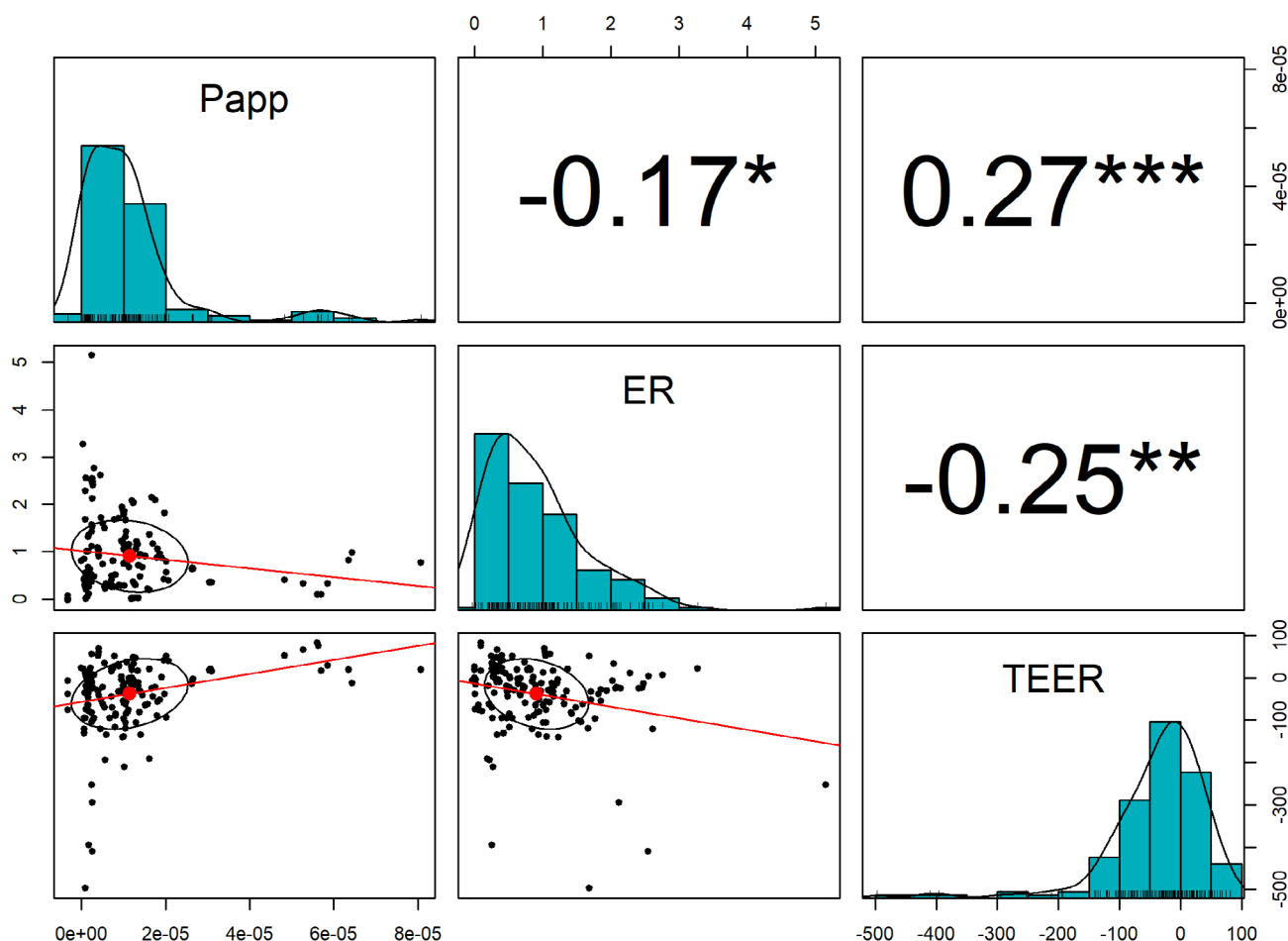


Figure 3. Pearson correlation coefficient (PCC) analysis for permeability indicators: Correlation matrix of permeability-related parameters across all phytochemicals, displaying density plots, 95% confidence ellipses and linear regression (LM) lines in scatter plots to better illustrate direct relationships. Statistical significance: * $P < 0.05$; ** $P < 0.01$ and *** $P < 0.001$.

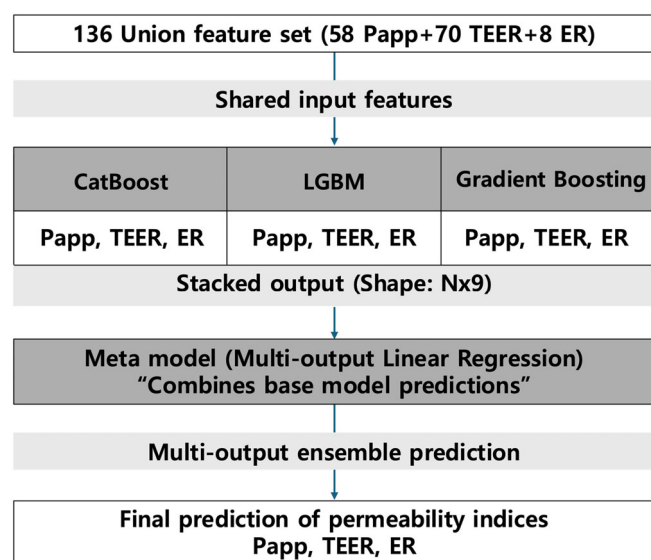


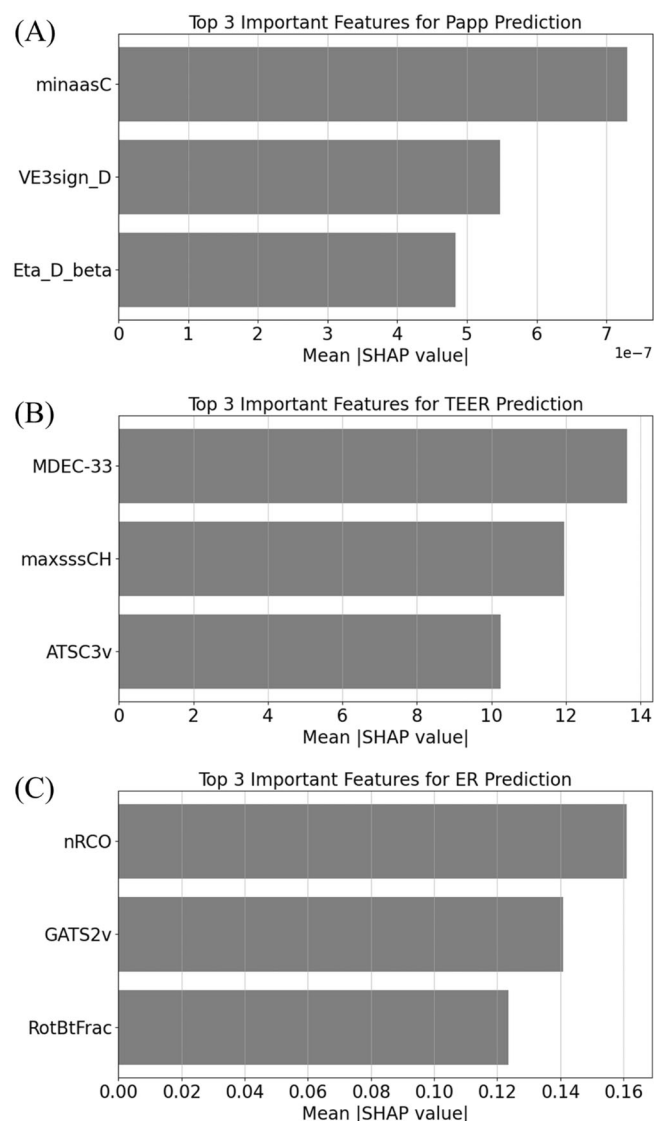
Figure 4. Stacking CatBoost-LGBM-gradient boosting ensemble model architecture for Papp, TEER and ER prediction.

Waals volume) and AATSC0p (Broto-Moreau autocorrelation weighted by polarizability), illustrated that subtle variations in molecular heterogeneity and polarity enhanced or restricted

passive diffusion across lipid bilayers. Overall, the selected descriptors demonstrated that Papp is a multifactorial phenomenon influenced not only by molecular size, but also by a complex interplay of structural topology, electronic distribution, charge localization and mass-weighted atomic properties. In the TEER prediction model, features such as MDEC-33, maxsssCH and ATSC3v, derived from 70 selected molecular descriptors, reflected key structural, topological and electronic properties associated with TEER prediction. Among these, the most influential descriptors identified through SHAP analysis were MDEC-33, maxsssCH and ATSC3v (Fig. 5(B)). These features collectively highlighted the critical importance of molecular branching, carbon connectivity and the spatial distribution of physicochemical properties in modulating TEER. In detail, MDEC-33, which represented the molecular distance edge between all tertiary carbons, was the most important feature. This descriptor suggested that the presence and spatial arrangement of tertiary carbon atoms, associated with molecular branching and bulkiness, critically influenced the tightness of epithelial cell monolayers. Increased branching could disrupt or loosen tight junctions, thereby modulating barrier integrity. The second most important feature, maxsssCH, denoted the maximum value of sssCH fragments, which reflected the prevalence of fully substituted carbon centers within a molecule. A high value of maxsssCH was indicative of highly substituted molecular structures, potentially affecting membrane interactions and tight junction modulation because

Table 2. Stacking ensemble model performance for bioavailability indicators.

Bioavailability indicators	NRMSE _{Train}	R ² _{Train}	NRMSE _{Test}	R ² _{Test}	Hyperparameters
Papp	0.0192 ± 0.0028	0.9862 ± 0.0030	0.0312 ± 0.0103	0.9550 ± 0.0399	Default settings random_state = 42
TEER	0.0493 ± 0.0091	0.8764 ± 0.0365	0.0781 ± 0.0316	0.5435 ± 0.4556	
ER	0.0334 ± 0.0017	0.9821 ± 0.0021	0.0593 ± 0.0181	0.9289 ± 0.0602	

**Figure 5.** SHAP-based feature importance profiles for the prediction of permeability indicators. The top descriptors visualized were selected from a final feature set comprising 58 descriptors for Papp (A), 70 for TEER (B) and 8 for ER (C).

of steric hindrance effects. The third most important feature, ATSC3v, represented centered Broto-Moreau autocorrelations weighted by van der Waals volumes. This descriptor highlighted the significance of atomic size distribution across molecular structures, which may influence membrane fluidity and tight junction behavior, ultimately affecting TEER values. Beyond these top-ranked features (Table S2), many of the remaining selected descriptors were also associated with autocorrelation functions

(MATS, GATS, ATSC and AATSC series), capturing the distribution of electronic, mass and charge properties within molecular frameworks. For example, AATS4v, a Broto-Moreau autocorrelation weighted by van der Waals volume, reflected how spatial variations in atomic size influence molecular packing and membrane interactions. In addition to these autocorrelation-based descriptors, SsssCH, representing the sum of E-states for fully substituted carbon centers, highlighted the contribution of highly branched carbon structures to the modulation of tight junctions through steric effects. Furthermore, shape-related indices such as path/walk indices (PW4 and PW5) and eigenvalue-derived features (SpDiam_B(v), SpMax_X) reflected aspects of molecular compactness, symmetry and overall spatial organization, all of which are critical determinants of epithelial barrier function. Collectively, these findings indicated that TEER was influenced by a complex interplay between molecular size, degree of branching, electronic distribution and structural compactness. In the ER prediction model, the most influential descriptor was the number of aliphatic ketones (nRCO), which captured the presence of polar functional groups known to significantly affect membrane permeability and efflux mechanisms (Fig. 5(C)). A higher number of ketone groups likely enhanced compound recognition by efflux transporters or altered passive diffusion characteristics. GATS2v (Geary autocorrelation of lag 2 weighted by van der Waals volume) emerged as another key feature, highlighting how spatial distribution of atomic volume influenced molecular interaction with efflux systems. RotBtFrac, representing the fraction of rotatable bonds, indicated that molecular flexibility contributed to efflux susceptibility by influencing conformational adaptability during transporter binding. In addition to these top-ranked descriptors, several other features provided complementary insights (see Supporting information, Table S2). MATS5s and AATSC6e are autocorrelation-based descriptors that capture the spatial distribution of ionization potential and electronegativity. ATSC5e similarly reflects electronic distribution patterns based on Sanderson electronegativity. VE1sign_Dz(Z) quantifies molecular topology via eigenvectors derived from atomic number-weighted adjacency matrices. LOC, the lopping centric index, describes the degree of molecular branching, which may affect membrane permeability through structural compactness and flexibility. Collectively, these findings demonstrated that ER prediction was not solely governed by specific functional groups, but rather shaped by an intricate interplay of molecular polarity, flexibility, electronic topology and structural complexity. Previous studies reported that permeability prediction was relied on lipophilicity-related descriptors such as CLogP, ALogP and KLogP.⁴⁷ These descriptors estimate the partitioning behavior of compounds between aqueous and lipid phases, which is a known determinant of passive permeability. However, such simplified metrics may not fully capture the structural and electronic complexity of diverse molecules, particularly phytochemicals with multiple functional groups and conformational flexibility. By contrast, the present study

employed a broader array of molecular descriptors, including E-state indices, spectral moments, autocorrelation functions and eigenvalue-based topological parameters. These descriptors captured nuanced features such as atomic electronic environments (minasC and VE3sign_D), molecular branching (MDEC-33 and LOC) and polarity distribution (GATS2v and AATSC4e). The SHAP analysis revealed that these factors collectively contributed to the prediction of Papp, TEER and ER with greater resolution. Compared to simple lipophilicity estimates, this multi-descriptor approach allowed for more mechanistic insights and robust performance across all three permeability indicators.

DISCUSSION

The development of explainable machine learning models for predicting permeability-related bioavailability indicators, Papp, TEER and ER, demonstrated significant advancements over traditional QSPR approaches. The regression models individually optimized for each bioavailability indicators, Papp, TEER and ER, using RFECV demonstrated superior predictive performance compared to those reported in previous studies.^{35,37} For example, a QSPR model was developed for Caco-2 cell permeability prediction using a combination of hybrid quantum particle swarm optimization (HQPSO) and dual-RBF neural networks, achieving an R^2 value of 0.85.³⁷ By contrast, the regression model developed for Papp prediction in this study achieved an R^2 of 0.9577, demonstrating markedly improved predictive performance compared to previously reported models. Similarly, previous study constructed QSPR models for natural products, reporting R^2 values around 0.74 for their best-performing models.³⁵ Models developed in the present study surpassed these benchmarks. The integration of SHAP for feature importance analysis provided valuable insights into the molecular descriptors influencing each bioavailability indicator. This interpretability aligns with the recommendations by Lundberg & Lee,⁴⁸ who emphasized the importance of model transparency in machine learning applications within the biomedical domain. By identifying key features contributing to model predictions, researchers can better understand the underlying biological mechanisms and guide future compound optimization efforts. Furthermore, the study's focus on natural phytochemicals expands the applicability of QSPR models beyond synthetic compounds, addressing a gap in the previous study. Previous models often relied on datasets comprising primarily synthetic molecules, limiting their relevance to natural products. By incorporating a diverse range of phytochemicals, the current models offer broader utility in natural product research and drug discovery. In summary, separate machine learning models were developed for each bioavailability indicator, Papp, TEER and ER, and integrated with SHAP-based interpretability to establish a robust and explainable framework for prediction. Based on the results of PCC analysis, which revealed interrelationships among three indicators, we hypothesized that a multi-output ensemble approach could exploit these correlations to enhance predictive performance. To test this, a stacking ensemble model was constructed by integrating predictions from base learners trained on a unified feature set. The ensemble model achieved high predictive performance for Papp and ER, likely reflecting the relatively direct and consistent physicochemical relationship between molecular features and the mechanisms governing membrane permeability and efflux. By contrast, TEER prediction exhibited substantially lower and more variable performance, suggesting that the biological complexity of tight junction

dynamics and ionic regulation, comprising factors poorly captured by structural descriptors, limited the utility of the ensemble approach for this endpoint. These findings underscore both the potential and limitations of multi-output ensemble modeling in the context of bioavailability prediction. They further emphasize the need for task-specific modeling strategies, particularly for biologically intricate indicators such as TEER. Despite the success of the ensemble model for Papp and ER, the high variability in TEER prediction highlights the limitations of relying solely on 2D and 3D molecular descriptors to model biologically intricate endpoints. TEER is deeply governed by dynamic cellular responses, including tight junction modulation and membrane fluidity. Future modeling strategies could significantly benefit from a multi-modal or hybrid feature approach. Incorporating features related to empirical cellular responses, such as specific tight junction protein expression data or measurements of monolayer integrity under different conditions, alongside advanced *in silico* metrics (e.g. specific binding affinities from molecular docking or membrane interaction properties) would likely capture the underlying biological factors influencing cellular integrity. This integrative approach would effectively complement the molecular descriptor approach, thereby improving the robustness and applicability of TEER predictions. Accordingly, the independently optimized models for each endpoint were prioritized, yielding not only reliable predictive performance, but also enhanced interpretability through SHAP-based feature importance analysis. The superior performance of these models relative to previously reported studies highlights their utility as valuable tools for the early-stage screening of natural compounds. In practice, this QSPR framework can be seamlessly integrated into existing drug discovery workflows as a high-throughput, first-tier screening filter. Phytochemicals predicted to possess optimal permeability profiles (e.g. high Papp and low ER) can be systematically prioritized for targeted experimental validation. Such downstream validations would logically include comprehensive *in vivo* pharmacokinetic profiling and advanced intestinal models to elucidate actual systemic absorption and therapeutic efficacy. However, it is critical to acknowledge the limitations of this approach. Although the developed QSPR models accurately predict Caco-2 cell permeability, the *in vitro* Caco-2 model itself cannot fully replicate the physiological complexity of the human gastrointestinal tract, lacking factors such as the intestinal microbiome, the mucus layer and dynamic food matrix interactions. Therefore, although these computational models significantly narrow down the chemical space and reduce the initial experimental burden, their predictions must ultimately be integrated with comprehensive *in vitro* and *in vivo* studies to fully confirm the systemic bioavailability and therapeutic efficacy of phytochemicals.

CONCLUSIONS

In the present study, we developed and validated machine learning models for predicting permeability-associated bioavailability indicators (Papp, TEER and ER) using molecular descriptors derived from 83 phytochemicals. Separate regression models were independently optimized for each target variable, yielding strong predictive performance. SHAP-based interpretation further provided mechanistic and chemical insights by identifying the most influential molecular descriptors contributing to each prediction endpoint. This modeling strategy not only improves the accuracy and interpretability of QSPR models, but also supports the early-stage screening of phytochemicals based on their absorption potential.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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